

Lab Exercise 12

SIMPLE PROGRAMMING USING WHILE & REPEAT

AIM

To implement **WHILE** and **REPEAT** loop statements in MySQL using the Vegetable database.

PROCEDURE

1. Select the VegetableDB database.
2. Create a function using the **WHILE** loop to repeatedly build a string.
3. Execute the function and display the result.
4. Create a function using the **REPEAT** loop to increase the vegetable price until it reaches ₹400.
5. Execute the function and display the final value.

Program 1: WHILE Loop

Task: Build a string repeatedly until x becomes greater than 5.

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```
USE VegetableDB;

DELIMITER //

CREATE FUNCTION VegetableWhile()
RETURNS VARCHAR(100)
DETERMINISTIC
BEGIN
    DECLARE x INT DEFAULT 1;
    DECLARE result VARCHAR(100) DEFAULT '';

    WHILE x <= 5 DO
        SET result = CONCAT(result, 'Vegetable ', x, ' ');
        SET x = x + 1;
    END WHILE;

    RETURN result;
END //

DELIMITER ;
```

Execute

```
SELECT VegetableWhile();
```

```
+-----+  
| VegetableWhile() |  
+-----+  
| Vegetable 1 Vegetable 2 Vegetable 3 Vegetable 4 Vegetable 5 |  
+-----+  
1 row in set (0.00 sec)
```

Program 2: REPEAT Loop

Since your table contains **vegetable prices**, we'll use `income` as a **price variable** and increase it until it reaches **4000**.

```
USE VegetableDB;  
  
DELIMITER //  
  
CREATE FUNCTION VegetableRepeat()  
RETURNS INT  
DETERMINISTIC  
BEGIN  
    DECLARE income INT DEFAULT 1000;  
  
    REPEAT  
        SET income = income + 500;  
    UNTIL income >= 4000  
    END REPEAT;  
  
    RETURN income;  
END //  
  
DELIMITER ;
```

Execute

```
SELECT VegetableRepeat();
```

```
+-----+  
| VegetableRepeat() |  
+-----+  
|                4000 |  
+-----+  
1 row in set (0.00 sec)
```